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A Hybrid Ensemble Approach with Feature Selection and Interpretability to Optimize Osteoporosis Risk Prediction

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Abstract— Osteoporosis is a prevalent disease characterized by reduced bone density and an increased susceptibility to fractures. Early detection is crucial for effective intervention. This study used a large dataset to construct a machine learning based framework for osteoporosis risk prediction. The preprocessing pipeline includes SMOTE normalization, scaling and class imbalance correction, followed by mutual information feature selection. On an 80:20 train-test split, several models and apply ensemble technique were trained and assessed, including random forest (RF), support vector machine (SVM), adaboost (ABC), extra trees classifier (ETC), decision tree (DTC) and XGBoost. Metrics like accuracy, precision, recall, F1-score and AUC score were used to evaluate the models. Ensemble of random forest and gradient boosting outperformed the other models, demonstrating its robustness in predicting osteoporosis risk with an AUC score of 83.05%. The model's interpretability was further improved by the incorporation of SHAP analysis which shed light on the contributions of important elements. This framework provides scalable solution for wider healthcare applications by showcasing the potential of machine learning to enhance osteoporosis prediction and help in making clinical decisions.

Keywords— Osteoporosis detection, SMOTE, machine learning, SHAP, Ensemble model, random forest and gradient boosting.

I. INTRODUCTION

Osteoporosis is a long term, progressive skeletal disease that causes bone tissue to deteriorate structurally and lose bone mineral density (BMD), making the bone more brittle and breaking more easily [1]. Osteoporosis is a major global health concern that impacts millions of individuals and is becoming more common as the population ages. The impact of hormonal changes make it most frequent in postmenopausal women but it can also affect males and younger people with chronic illnesses such rheumatoid arthritis, chronic kidney disease or long-term corticosteroid usage [2]. Osteoporotic fractures, particularly those affecting the hip, spine, and wrist, are among the leading causes of disability, reduced mobility, diminished quality of life, and an increased risk of mortality in older adults, sometimes referred to as a "silent disease" develops gradually and shows no symptoms until a fracture happens [3]. This emphasizes how crucial early detection and preventative.

Measures are Age, hormone imbalances, genetic predisposition, poor diet (especially calcium and vitamin D deficits), inactivity, excessive alcohol consumption, smoking, concomitant conditions like diabetes and inflammatory disorders are risk factors for osteoporosis [4]. Bone densitometry, clinical evaluations, and prediction techniques such as the FRAX model which integrates clinical risk factors with bone mineral density to estimate fracture risk are commonly used to establish the diagnosis. Because osteoporosis is asymptomatic and fracture risk prediction is complicated, controlling osteoporosis remains difficult even with improvements in diagnostic techniques and pharmaceutical treatments. Machine learning(ML) and artificial intelligence(AI) have recently emerged as novel ways to improve the accuracy of risk assessments and identifying high risk individuals earlier [5]. These computational tools use massive datasets such as clinical records and lifestyle characteristics to create personalized preventative and treatment solutions. As the worldwide burden of osteoporosis grows, utilizing these technologies presents great opportunities to enhance outcomes while lowering healthcare costs. Make points below the contributions of our work.

- Suggested a scalable and methodical methodology for predicting different medical diseases that may be used with comparable datasets.
- Used SMOTE to address class imbalance in osteoporosis cases, ensuring strong model performance for predictions of minority classes.
- Created and contrasted several ensemble machine learning models (such as RF, SVM and XGBoost) to determine the method for predicting osteoporosis.
- Interpreted feature contributions using SHAP values, improving prediction transparency and reliability.

II. RELATED WORK

Yi-ting Lin et al. [6] used optimized feature selection and retrospective clinical data to compare four machine learning models: ANN, RF, SVM and LR. Although overall performance was comparable, ANN outperformed LR in severe situations (T-score < -2.5) and LR excelled in recall for general cases. ChaeWon Lee et al. [7] using data from 2,374 individuals

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in the KORONA cohort, this study created machine learning models for predicting osteoporosis, identifying major risk factors such as BMI, age and menopause. Logistic regression showed the highest AUC (0.750) whereas XGBoost had the highest accuracy (0.682), highlighting ML's promise in RA related osteoporosis screening. Jun-bo Tu et al. [8] a machine learning model was developed to predict osteoporosis using data from 10,000 patients in the German Disease Analyzer database. The stacker model, which combined logistic regression, AdaBoost, and gradient boosting, demonstrated the highest performance in identifying key factors such as age, gender, lipid disorders, cancer, and COPD. The approach has great potential for early identification and tailored osteoporosis care.

Alexander Engels et al. [9] compared the superlearner method to established methods in predicting osteoporotic hip fractures using claims data from 288,086 persons aged 65 and higher. Support vector machines and RUSBoost algorithms produced AUCs of 0.66-0.72 for training and 0.65-0.70 for validation. The superlearner performed well in training but struggled with discrimination and calibration during validation. Jae-geum Shim et al. [10] used KNHANES data to create machine learning models were used to predict osteoporosis in Korean postmenopausal women. Among seven methods, ANN had the highest AUROC (0.743) with feature selected data, demonstrating the importance of streamlined inputs in accurate prediction.

ILIOU et al. [11] assessed 20 machine learning models for predicting osteoporosis in 3,426 people based on age, gender, height and weight. Feature selection demonstrated comparable accuracy with only age and weight, implying a simpler method to diagnosis and reduction in bone densitometry utilization. Kainat A. Ullah et al. [12] using data from Taiwan and Korea, this study created a machine learning model that outperformed conventional techniques in predicting osteoporosis by utilizing ANN, SVM and KNN. Early detection is made possible by the system which helps with proactive management and difficulties avoidance. German Cuaya-Simbro et al. [13] found that random forest and IBk (KNN) classifiers with oversampling were effective in predicting falls in 126 older women with osteoporosis using balance measures. Improved fall prediction thanks to key balance factors facilitates clinical evaluation. Su Jeong Kang et al. [14] demonstrated the potential of machine learning for better risk assessment by finding that gradient boosting (AUROC 0.662) performed better in predicting osteoporotic fractures than the FRAX score (AUROC 0.617).

III. Methodology

In **Fig. 1** the process started with preprocessing the dataset which included normalization and scaling to normalize features and the use of SMOTE to address class imbalance and ensure the equal representation of instances of osteoporosis. The most relevant predictors were found through the use of mutual information in feature selection which improved model efficiency and decreased complexity. Ensemble machine learning techniques, including random forest, SVM, XGBoost, adaboost, decision tree and extra tree were trained to predict osteoporosis after the data was divided into training (80%) and testing (20%) subsets. SHAP (shapley additive explanations) values were computed to improve interpretability and provide information on feature contributions to predictions. The accuracy, precision, recall, F1-score and AUC metrics were used to thoroughly assess the model's performance. In addition to ensuring balanced, optimized and interpretable models, this all-encompassing strategy showed promise for reliable osteoporosis risk prediction which successfully supports clinical decision making.

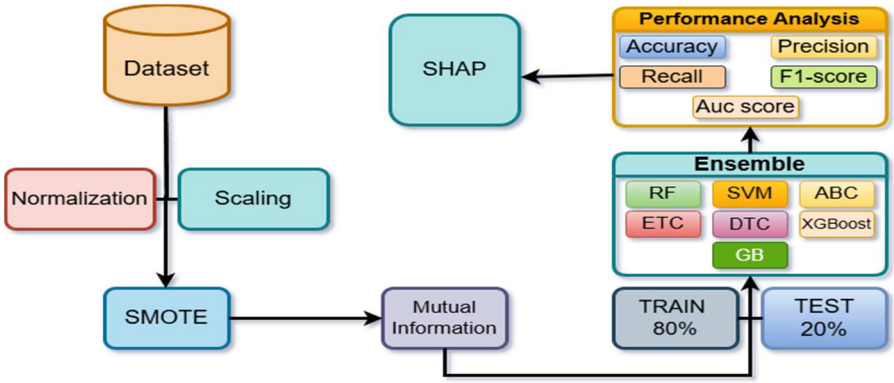


Fig. 1: Workflow of the proposed model

A. Dataset

This dataset is collected from primary healthcare data from the german analyzer database (IMS HEALTH) [15] and provides information on 10,000 patients, including their age and gender as well as the presence or absence of ten chronic diseases such as hypertension, diabetes, cancer and chronic kidney disease. Each patient is recognized uniquely and all conditions are represented as binary variables (1=present, 0=absent) allowing for simple prevalence and comorbidity analysis. In the dataset, there are no missing values, the information is full and may be used to identify high risk groups, investigate the connections between age, gender and disease trends and support epidemiological research and healthcare resource planning.

B. Data processing

Preprocessing the data made sure it was clean and prepared for modeling. SMOTE was utilized to create synthetic samples for the minority class in order to address class imbalance (osteoporosis cases) while continuous variables were standardized and scaled to standardize feature ranges. There are no missing values in this dataset and superfluous features have been eliminated for increased efficiency. By creating a balanced, uniform dataset, these procedures improved model performance and ensured accurate predictions.

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C. Feature Selection

Selecting features based on mutual information determines which traits are most pertinent by calculating how dependent they are on the target variable, taking into account both linear and nonlinear correlations. This method reduced dimensionality, improved model efficiency and accuracy and allowed the elimination of less useful variables by ranking features according to their importance [16]. Seven features selected based on mutual information score in **TABLE-I**.

TABLE-I: FEATURE SELECTION BY MUTUAL INFORMATION SCORE

Feature	Mutual Information Score	Selected
male_gender	0.110640	True
age	0.032486	True
Hypertension	0.009304	True
Diabetes	0.005957	True
Heart_failure	0.005249	True
COPD	0.003549	True
CHD	0.003117	True
Chronic_kidney_disease	0.001788	False
Lipid_disorder	0.001131	False
Stroke	0.000258	False

D. Data Split

To enhance model assessment, the dataset is divided into training and testing portions, assigning 80% to training while reserving 20% for evaluation. This ensures that models learn from a subset of the data while being tested on previously unseen data to test generalization and avoid overfitting. The train_test_split function from sklearn is used for this with features (X) and the target variable (y) separated and stratified sampling to ensure the target class distribution remains consistent across subsets.

E. Random Forest

It is a strong and adaptable ensemble machine learning approach that is mostly applied to regression and classification problems. In order to produce more precise and reliable predictions, it builds a large number of decision trees during training [17]. Random forest is a popular option for many machines learning tasks because of its capacity to manage big datasets and avoid overfitting. In addition, they offer feature importance scores which aid in comprehending the contribution of each feature to the prediction, thus providing both strong performance and interpretability. The contamination is present by the P_{ij}

$$P_{ij} = L_j N_j - L_{left(j)} N_{left(j)} - L_{right(j)} N_{right(j)} \quad (1)$$

F. Gradient Boosting

It produces models in an orderly way with the goal of fixing the mistakes produced by the models before it [18]. It is recognized as well for being precise and efficient, particularly in activities like Kaggle. A few well-liked implementations are LightGBM, CatBoost and XGBoost.

IV. RESULT AND DISCUSSION

Accuracy examines overall correctness, precision indicates positive prediction accuracy and recall evaluates the ability to detect true positives. The F1 Score balances precision and recall whereas ROC-AUC assesses classification performance across many thresholds [19, 20]. High performance CPUs or GPUs are used by the framework to handle massive datasets and train intricate machine learning models. We determined the performance parameter by using the following equations: (2), (3), (4), (5) and (6).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

$$Sensitivity = \frac{TN}{TN + FP} \quad (4)$$

$$F_1 - Score = 2 * \left(\frac{Sensitivity * Precision}{Sensitivity + Precision}\right) \quad (5)$$

$$Auc = \frac{1}{2} * (Sensitivity + SpecificitY) \quad (6)$$

A. Confusion Metrics

In **Fig. 2** the confusion matrices compare different models: RF(a), ETC(b), SVM(c), AB(d), RF+GB(e) and RF+XGB(f). Among them, RF+GB(e) performed the best, demonstrating its outstanding predictive capacity with 1,208 true positives (TP), 912 true negatives (TN) and the fewest false negatives (156) and false positives (518). RF+XGB(f) came in second with low errors, 934 TN and 1,195 TP. Despite having a high number of true positives (1,222), SVM(c) had trouble with false positives (690). Similar results were obtained by ETC(b) and RF(a) with balanced predictions but marginally greater false positives and false negatives. The precision of AB(d) was the lowest with 593 false positives. To predicting osteoporosis, RF+GB(e) performed the best overall with RF+XGB(f) coming in second.

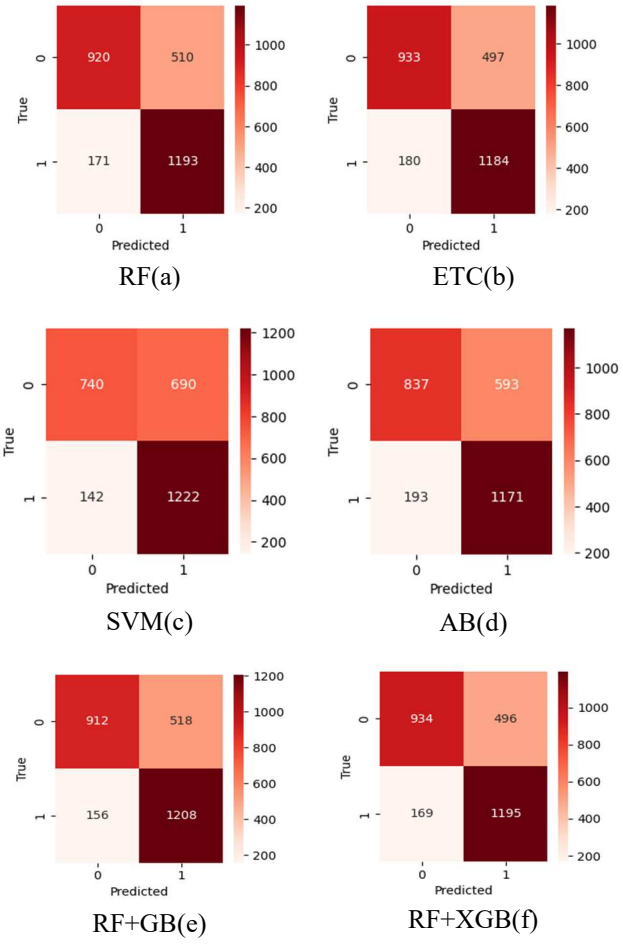


Fig. 2: Confusion metrics for ML and Ensemble.

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B. *Roc Curve*

In **Fig. 3** ROC curve shows well the random forest model predicts osteoporosis risk. The curve has an AUC score of 0.759, indicating high discriminatory ability. The significant deviation from the diagonal random guess line demonstrates that the model is successful at discriminating between positive and negative classes.

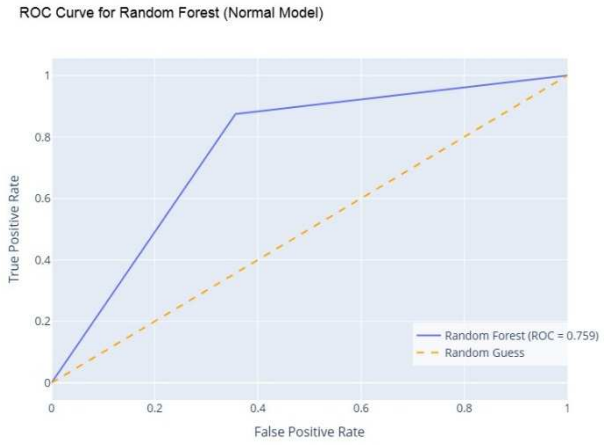


Fig. 3: Random Forest Roc Curve.

In **Fig. 4** the ROC curve shows the combined performance of combined the random forest (RF) and gradient boosting (GB) models in predicting osteoporosis risk. The improved predictive capability through model integration is indicated by the AUC score of 0.762 which shows a modest improvement over utilizing random forest alone **Fig. 3**. The curve's noticeable divergence from the diagonal random guess line demonstrates that the model can differentiate between classes. Furthermore, a more robust and trustworthy prediction method for osteoporosis risk assessment is ensured by RF and GB cooperative approach.

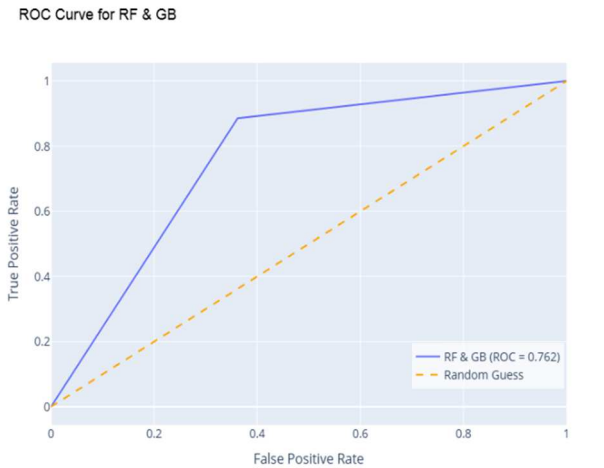


Fig. 4: (RF+GB) Roc Curve.

C. *Figures and Tables*

In **TABLE-II**, XGB performed the best with a ROC of 76.24%, accuracy of 75.98% and recall of 87.10%. ETC and RF had accuracies of 75.77% and 75.63%, respectively and RF had the highest AUC (82.85%). Although SVM's accuracy was the lowest at 70.22%, its recall was the best at 89.59%. Overall, the models for predicting osteoporosis were XGB, ETC and RF are good.

TABLE-II: MACHINE LEARNING MODEL PARAMETERS SCORE WITH FEATURE SELECTION

Model	Accuracy	Precision	Recall	F1	ROC	AUC
RF	75.63	70.05	87.46	77.80	75.90	82.85
SVM	70.22	63.91	89.59	74.60	70.67	77.43
AdaBoost	71.87	66.38	85.85	74.87	72.19	78.54
ETC	75.77	70.43	86.80	77.77	76.02	82.69
DTC	75.70	70.30	86.95	77.75	75.96	82.28
XGB	75.98	70.59	87.10	77.98	76.24	82.68
GB	71.62	66.72	85.54	74.97	71.71	78.53

In **TABLE-III**, the best performing voting-based model was RF+GB (random forest + gradient boosting) which had the highest AUC (83.05%) and good ROC (76.17%). Following closely behind with high accuracies (75.88% and 76.20%, respectively) and AUC values of 83.04% were ET+GB and RF+XGB. Even though RF+ADA and ET+XGB had somewhat worse results, their AUCs of 82.94% were still high. overall, RF+GB showed the best accuracy and expressed power balancing.

TABLE-III: APPLY VOTING ON VARIOUS MODEL WITH FEATURE SELECTION

Model	Accuracy	Precision	Recall	F1	ROC	AUC
RF+GB	75.88	69.99	88.56	78.19	76.17	83.05
ET+GB	75.88	69.97	88.64	78.20	76.17	83.04
RF+XGB	76.20	70.67	87.61	78.23	76.46	83.04
ET+XGB	76.16	70.70	87.39	78.16	76.42	82.94
RF+ADA	75.73	70.32	87.02	77.79	75.99	82.94

In **Fig. 5** the accuracy percentage of ML models for osteoporosis prediction following feature selection is contrasted with the mutual information classifier in this bar chart. Decision tree (DT) had the highest accuracy (74.60%), just exceeding random forest (RF) (74.57%). Other high performing models were XGBoost (74.53%) and extra trees model (74.46%). Gradient boosting (71.62%) and adaboost (71.59%) demonstrated intermediate accuracy whereas SVM had the lowest accuracy (70.87%).

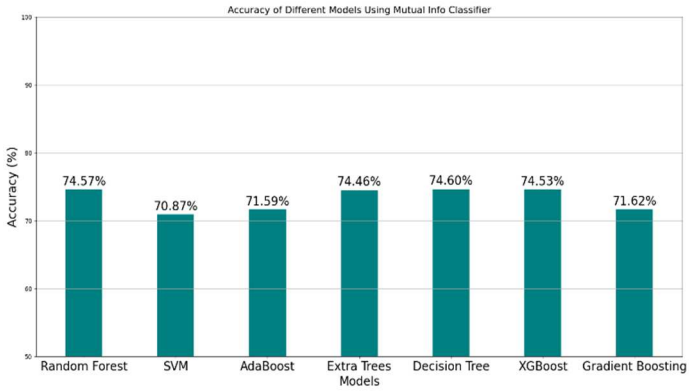


Fig. 5: Seven models accuracy score.

In **Fig. 6** the accuracy of different ensemble model combinations for osteoporosis prediction is shown in the bar chart. The best accuracy was 76.20% for RF+XGB (random forest + XGBoost) which was closely followed by 76.16% for ET+XGB (extra trees + XGBoost). The accuracy of RF+ADA (random forest + AdaBoost) was 75.73%, while that of RF+GB (random forest + gradient boosting) and ET_GB (extra trees + gradient boosting) was 75.88%. Overall, the combination of RF+GB is good for other performance which represent in **TABLE-III**.

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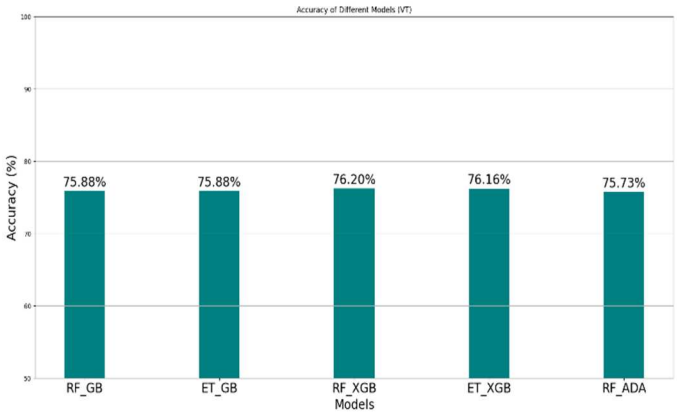


Fig. 6: Top Five accuracy score in voting.

In **TABLE-IV**, our method effectively overcame the limitations of earlier research. Because of inadequate preprocessing and model adjustment, Yi-Ting Lin et al. [6] and Alexander Engels et al. [9] reported AUC scores of 73.1% and 72.0%, respectively. Despite achieving 75.0% and 75.3%, respectively, ChaeWon Lee et al. [7] and Jun-bo Tu et al. [8] models lacked a strong class imbalance adjustment. Our approach addressed these shortcomings using SMOTE, mutual information-based feature selection and sophisticated ensemble models random forest and gradient boosting, resulting in a higher AUC of 83.05%. Our method for clinical osteoporosis prediction is more efficient and scalable. SHAP analysis which also improved interpretability.

TABLE-IV: PERFORMANCE ANALYSIS WITH OTHER WORKS

Reference	Dataset	AUC Score (%)
Yi-ting Lin et al. [6]	Data collected from EMR	73.1
ChaeWon Lee et al. [7]	Data collected from KORONA 2374 patients	75.0
Jun-bo Tu et al. [8]	Data collected from IMS HEALTH 10,000 patients	75.3
Alexander Engels et al. [9]	288,086 patient's dataset	72.0
Proposed work	**Data collected from IMS HEALTH 10,000 patients**	**83.05**

D. Interpretability

In **Fig. 7** the contributions of distinct features to a particular model prediction are shown in this SHAP plot. Features like age, heart failure and coronary heart disease (CHD) have a negative influence (in blue) which lowers the likelihood of a positive outcome whereas features like male gender and diabetes have the biggest positive impact (in red), greatly boosting the odds of a positive prediction. Both COPD and hypertension show marginal contributions with the latter somewhat lowering the prediction likelihood and the former slightly raising it.

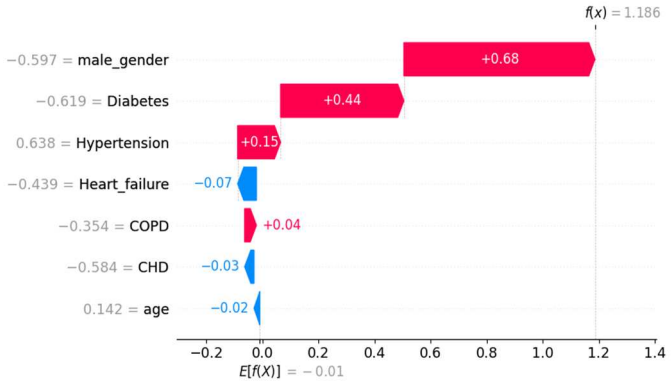


Fig. 7: SHAP Analysis Key Features Contributing to Model Predictions.

In **Fig. 8** SHAP force graphs explain individual predictions by displaying feature contributions with red segments pushing predictions into the positive class and blue segments pulling them into the negative. Age, male gender and hypertension are all major predictors of good outcomes whereas CHD, COPD and heart failure frequently counterbalance.



Fig. 8: SHAP Force plots for individual predictions.

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V. Conclusion

This study uses an efficient and comprehensible methodology to demonstrate the promise of machine learning in osteoporosis prediction. The models were adapted to efficiently capture the patterns linked to osteoporosis risk by resolving class imbalance with SMOTE and choosing pertinent features through mutual information. Strong prediction skills were shown by ensemble machine learning algorithms like Random Forest, SVM, and XGBoost, gradient boosting while SHAP analysis improved model interpretability by shedding light on the significance of important characteristics. A thorough assessment of the model's performance demonstrated their potential as useful instruments for risk classification and early identification, confirming their dependability. In addition to enhancing osteoporosis prediction, the suggested methodology offers a scalable and effective framework that may be used as a model for implementing machine learning to address other healthcare issues.

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